

Data Analyst & AI Engineer

Professional Summary

Data Analyst & AI Engineer with strong expertise in **machine learning**, **data visualization**, and **full-stack analytics solutions**. Skilled in transforming raw data into actionable insights through tools like **Python**, **Power BI**, and **SQL**, and experienced in developing and deploying **AI/ML models** using **TensorFlow**, **Scikit-learn**, and **Streamlit**.

Adept at building **data-driven dashboards**, **predictive analytics systems**, and **automation workflows** that improve business decision-making and operational efficiency. Passionate about leveraging **AI innovation and data science** to drive measurable value across business and engineering environments.

Professional Experience

FlyRank | Machine Learning Internship | Egypt | Jul 2025 – Present

Developing and optimizing machine learning models and data pipelines to support AI-driven product features, applying Python, Scikit-learn, and deep learning techniques to real-world datasets.

AITB | Data Analyst Internship | Egypt | Jul 2025 – Sep 2025

Analyzed business datasets to extract actionable insights, built interactive dashboards and reports using Python and Excel, and supported data-driven decision-making across internal teams.

CodeAlpha | Internship | Egypt | Jul 2025 - Aug 2025

Executed 4 end-to-end analytics projects using real-world datasets to perform data collection, cleaning, visualization, and NLP analysis with Python and Jupyter Notebooks, uncovering actionable insights and improving data-driven decision support.

University Student | Future University in Cairo | 2022 – 2026

- Completed in-depth research into course topics to better understand the material.
- Met course goals and achieved good marks with diligent completion of assignments.
- Prepared for examinations by studying course material with fellow students.
- Studied different subjects by completing extra-curricular activities and courses.
- Helped complete and lead assigned group projects, consistently meeting objectives and deadlines.
- Followed along with lectures and made notes for later reference and study.
- Maintained a balance of education, sports, and volunteer activities as a student.
- Advanced class discussions by asking questions and suggesting answers to others' inquiries.
- Joined clubs to build relationships and connections with students.
- Organised courses and studied hard to prepare for a career in the field.
- Supported the community as a volunteer, helping meet important needs and expanding horizons.
- Played sports and trained consistently to improve skills and conditioning for challenging games.

Projects

EmpowerHR — AI-Powered HR Management System | Sep 2024 to Jul 2026

Key Technologies/Tools: Django · React.js · PostgreSQL · Docker · Railway · Cloudinary · Sentence-BERT · XGBoost

An AI-powered, cloud-based HR Management System designed for SMEs in Egypt, unifying the full employee lifecycle in one affordable platform. Features AI CV ranking (Sentence-BERT + TF-IDF cosine similarity), attrition prediction (XGBoost trained on IBM HR Analytics dataset), localized Egyptian salary benchmarking, geofencing-based attendance tracking, payroll, leave management, performance reviews, and task planning.

Live: <https://empower-hr-v2.vercel.app/> **GitHub:** <https://github.com/AlaaFarouk37/EmpowerHR-v2>

HR Analytics Dashboard | Jun 2025 to Aug 2025

Key Technologies/Tools: Python, Pandas, NumPy, Power BI, Node.js, Angular, Matplotlib, Seaborn.

Designed and deployed an interactive HR analytics dashboard using Python, Node.js, and Angular to identify key drivers of employee attrition from large HR datasets, improving retention analysis efficiency by 30% through automated insights visualization.

Heart Disease Risk Prediction | Jun 2025 to Aug 2025

Key Technologies/Tools: Python, Scikit-learn, Streamlit, Ngrok, Pandas, NumPy, Matplotlib

Developed a full machine learning pipeline using Python and Streamlit to predict heart disease risk with the UCI dataset, achieving 92% model accuracy after feature selection, model tuning, and deployment for real-time prediction access via Ngrok.

Hotel Monitoring Dashboard | Jul 2025 to Aug 2025

Key Technologies/Tools: Python, Streamlit, Pandas, Seaborn, Plotly, Excel

Built a hotel operations dashboard in Python and Streamlit to analyze booking trends, cancellations, and revenue KPIs; enabled data-driven revenue optimization and customer behavior insights through automated daily updates.

AI Data Clustering | Jan 2025 to Jun 2025

Key Technologies/Tools: R, ggplot2, tidyverse, dplyr, CSV datasets

Implemented K-Means and Hierarchical Clustering in R (tidyverse, ggplot2) to segment customer behavior datasets, identifying 3 high-value customer groups that guided targeted marketing recommendations.

Food Ordering Website | Sep 2024 to Feb 2025

Key Technologies/Tools: Python, TensorFlow, OpenCV, Flask, HTML/CSS, JavaScript

Engineered a web-based food ordering platform using Python, TensorFlow, and OpenCV with CNN-based image classification for menu recognition; enhanced order accuracy by 25% through AI-enabled food item detection.

Music-Player-Application | Feb 2023 to Jun 2023

Key Technologies/Tools: C++, OpenMP, Qt Framework, OOP, Git

Developed a C++ GUI music player using OpenMP for concurrent audio playback and playlist management, improving processing performance and user interactivity for large audio libraries.

Skills

Data Analysis & Visualization

- Data Cleaning, Exploratory Data Analysis (EDA), KPI Analysis, Time Series Forecasting
- Dashboarding & Reporting: Power BI, Streamlit, Matplotlib, Seaborn, Plotly, Excel
- Data Wrangling: Pandas, NumPy, SQL, R (tidyverse, ggplot2)

Machine Learning & Artificial Intelligence

- Supervised Learning: Logistic Regression, SVM, Random Forest, Gradient Boosting
- Unsupervised Learning: K-Means, Hierarchical Clustering, PCA, Anomaly Detection
- Deep Learning: CNN, RNN, Transformer-based models, TensorFlow, PyTorch
- Natural Language Processing (NLP): Sentiment Analysis, Text Classification, Tokenization, Vectorization
- Model Optimization: Feature Engineering, Hyperparameter Tuning, Cross-Validation

Cloud & Deployment

- Model Deployment: Streamlit, Flask, Docker, AWS, Azure, Ngrok
- CI/CD & Version Control: Git, GitHub, GitLab
- Containerization & Virtualization: Docker, Anaconda, Virtual Environments

Programming & Development

- Languages: Python, C++, Java, R, SQL, JavaScript, HTML/CSS
- Web Development: Flask, ASP.NET, Node.js, Angular
- Desktop Development: Java Swing, Qt Framework, C++ GUI
- Software Practices: Object-Oriented Programming (OOP), Debugging, Unit Testing

Data Engineering Tools

- Data Processing & ETL: BeautifulSoup, Requests, APIs, CSV/JSON handling
- Database Interaction: MySQL, SQLite, PostgreSQL

- Workflow Automation: Jupyter Notebooks, VS Code, PyCharm

Soft Skills

- Analytical Thinking | Problem Solving | Collaboration | Technical Documentation | Communication

Education

Bachelor of Computers and Information Systems | 2022-2026

Future University in Egypt (Cairo)

- CGPA: 3.4 / 4.00, First-Class Honors
- Key Courses: Machine Learning, Data Analysis, Artificial Intelligence, Databases, Software Engineering
- Graduation Project: EmpowerHR — AI-Powered HR Management System for SMEs (Flutter, Python, Sentence-BERT, XGBoost)

Certifications

- Data Analysis @AITB | 2025
- Sprints x Microsoft Summer Camp - AI and Machine Learning | 2025
- Data Analysis using Microsoft Excel | 2025
- Business Analysis & Process Management | 2025
- Microsoft Excel | 2025
- Cloud Basics @Huawei | 2025

Languages

- Arabic: Native
- English: Fluent